

Grade 11 lessons

Algebra 2: complex numbers, logs and sequences

IN THIS BOOK

1. Complex numbers
2. Exponents and logs
3. Sequences and series
4. Function moves
5. Polynomials and rational expressions

HOW TO USE IT

Read one lesson together, then practice it. Each lesson matches 5 questions in the Grade 11 Kahoot and Blooket sets, and one stop on the Grade 11 trail in the Math Trail app.

Every lesson has the big idea, the steps, two worked examples, one mistake to watch for, and the words to know.

Complex numbers

THE BIG IDEA

The number i is defined as the square root of -1 , so $i^2 = -1$. It lets us take square roots of negative numbers. A complex number has a real part and an i part, like $3 + 2i$.

HOW TO DO IT

1. $\sqrt{-25} = \sqrt{25} \times \sqrt{-1} = 5i$.
2. Powers of i repeat every four: i , -1 , $-i$, 1 . Divide the power by 4 and use the remainder.
3. To add, combine the real parts and combine the i parts.
4. To multiply, expand the brackets, then replace i^2 with -1 .
5. If the discriminant $b^2 - 4ac$ is negative, the quadratic has complex solutions.

WORKED EXAMPLES

$$(2 + 3i) + (1 - i) = ?$$

Real parts: $2 + 1 = 3$. i parts: $3i - i = 2i$.
Answer: $3 + 2i$.

$$(1 + 2i)(3 + i) = ?$$

Expand: $3 + i + 6i + 2i^2$. Replace i^2 with -1 : $3 + 7i - 2 = 1 + 7i$.

WATCH OUT

i^2 is -1 , so the last term changes sign and joins the real part. Forgetting this is the most common mistake.

WORDS TO KNOW

imaginary unit

the number i , where $i^2 = -1$

complex number

a number written as $a + bi$

Exponents and logs

THE BIG IDEA

A logarithm asks a question: what power do I need? $\log_2 8 = 3$ because $2^3 = 8$. Logs undo exponents.

HOW TO DO IT

1. To find log of a number, ask what power of the base gives that number.
2. To solve $3^x = 81$, write 81 as a power of 3.
3. A fraction exponent is a root: the bottom number is the root, the top is the power.
4. Log rules: $\log(ab) = \log a + \log b$, $\log(a \div b) = \log a - \log b$, $\log(a^n) = n \times \log a$.
5. For any base: the log of 1 is 0, and the log of the base itself is 1.

WORKED EXAMPLES

Solve $3^x = 81$.

$81 = 3 \times 3 \times 3 \times 3 = 3^4$. So $x = 4$.

$8^{2/3} = ?$

Cube root first: the cube root of 8 is 2.
Then square it: $2^2 = 4$.

WATCH OUT

$\log(a + b)$ is not $\log a + \log b$. The adding rule is for multiplying inside the log.

WORDS TO KNOW

logarithm

the power needed to reach a number

base

the number being raised to a power

Sequences and series

THE BIG IDEA

A sequence is a list of numbers that follows a rule. Arithmetic sequences add the same amount each time. Geometric sequences multiply by the same amount each time.

HOW TO DO IT

1. Arithmetic: find the common difference d by subtracting neighbors. Term $n = \text{first} + (n - 1) \times d$.
2. Geometric: find the common ratio r by dividing neighbors. Term $n = \text{first} \times r^{n-1}$.
3. Sum of an arithmetic sequence = number of terms $\times$ (first + last) $\div 2$.
4. A handy case: $1 + 2 + 3 + \dots + n = n(n + 1) \div 2$.

WORKED EXAMPLES

5, 8, 11, ... What is term 10?

The difference is 3. Term $10 = 5 + 9 \times 3 = 32$.

3, 6, 12, ... What is term 6?

The ratio is 2. Term $6 = 3 \times 2^5 = 3 \times 32 = 96$.

WATCH OUT

It takes $n - 1$ jumps to reach term n , not n jumps. Term 10 is 9 jumps from the first term.

WORDS TO KNOW

common difference

the amount added each time

common ratio

the amount multiplied each time

Function moves

THE BIG IDEA

Functions can be combined, reversed and moved around. Each change to the rule moves or reverses the graph in a predictable way.

HOW TO DO IT

1. $f(g(x))$ means do g first, then put that answer into f .
2. To find an inverse, swap x and y , then solve for y .
3. $y = f(x) + k$ moves the graph up k . $y = f(x) - k$ moves it down.
4. $y = f(x - h)$ moves the graph right h . $y = f(x + h)$ moves it left.
5. $y = (x - h)^2 + k$ is the basic parabola with its vertex moved to (h, k) .

WORKED EXAMPLES

$f(x) = 2x + 1$ and $g(x) = x^2$. Find $f(g(3))$.

$g(3) = 9$. Then $f(9) = 2 \times 9 + 1 = 19$.

Find the inverse of $f(x) = 3x + 6$.

Write $y = 3x + 6$. Swap: $x = 3y + 6$.

Solve: $y = (x - 6) \div 3$.

WATCH OUT

Changes inside the bracket work backwards. $x - 3$ moves the graph right, and $x + 3$ moves it left.

WORDS TO KNOW

composition

using the output of one function as the input of another

inverse

the function that undoes another

Polynomials and rational expressions

THE BIG IDEA

Higher-degree polynomials have zeros, remainders and end behavior that you can read from the way they are written.

HOW TO DO IT

1. Remainder theorem: the remainder when $p(x)$ is divided by $(x - a)$ is $p(a)$. Just plug in a .
2. Zeros: in factored form, each bracket gives one zero. $(x - 3)$ gives 3 and $(x + 4)$ gives -4.
3. To simplify a fraction of polynomials, factor the top and bottom, then cancel matching brackets.
4. End behavior: even degree means both ends go the same way. Odd degree means they go opposite ways. A positive leading number means the right end goes up.

WORKED EXAMPLES

Find the remainder when $x^3 - 2x + 5$ is divided by $(x - 2)$.

Plug in 2: $8 - 4 + 5 = 9$. The remainder is 9.

Simplify $(x^2 - 9) \div (x - 3)$.

Factor the top: $(x + 3)(x - 3)$. Cancel $(x - 3)$. Answer: $x + 3$.

WATCH OUT

The zero from $(x + 4)$ is -4, not 4. And only cancel whole brackets, never single terms.

WORDS TO KNOW

zero

an x value that makes the polynomial equal 0

degree

the highest power of x